The Galactic–Cosmological Nexus: Connecting CRM to MOND Dynamics via Curvature Saturation

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Papers I–III of the Curvature Relaxation Model (CRM) series demonstrated that a scalar sector ($R + \gamma R^2$ + Pöschl-Teller saturation) reproduces the cosmological dark sector ($\Delta\chi^2 = -5.5$ vs. ΛCDM, zero EDE, $\alpha_T = 0$). However, the scalar sector alone cannot produce flat rotation curves: the Yukawa fifth force yields at most $\mu \to 4/3$, insufficient to replace dark matter halos. This paper introduces a second degree of freedom – a timelike unit vector field A_μ – derived from thermodynamic necessity (maximum entropy production, hierarchical dissipation). The teleodynamic coupling $\mathcal{F} = |T|/\rho_{\rm crit} \cdot \mathrm{sech}^2(\phi/\phi_0) \cdot A_\mu \partial^\mu \phi$ ensures BBN protection, links MOND strength to expansion dynamics, and couples local structure to the cosmological expansion rate. The model yields $a_0 = cH_0/(2\pi)$ as a consistency relation – linking local galactic dynamics to the cosmological expansion rate – and reproduces the Radial Acceleration Relation and flat rotation curves without dark matter particles. We emphasise that this is not a parameter-free derivation: a_0 is fixed by matching the observed value, and the relation $a_0 = cH_0/(2\pi)$ constitutes a testable consistency check, not a proof. The 4/3 convergence factor from $f(R)$ perturbation theory (Paper III) coincides with the MOND phase-space ratio V_3/V_2 (Paper II). The complete action contains 8 terms; 5 are uniquely determined by axioms, 4 parameters follow from data. *Scope:* solar-system compatibility is *not* established. The screening previously invoked for it does not follow from the quadratic $f(R)$ sector of Paper III, where the corresponding claim has been withdrawn; the Newtonian–MOND transition is therefore reported here as a phenomenological result computed with an assumed density-dependent scalaron mass, while the mechanism that would supply that mass remains an open problem. No Λ, no dark matter particles – only geometry.

EXECUTIVE SUMMARY

This paper extends the Curvature Relaxation Model (CRM) from the cosmological (Papers I–II) and scalar-field (Paper III) scales to galactic dynamics. The central result is that the CRM's curvature-saturation mechanism, when coupled to a massive vector field, reproduces the empirical MOND phenomenology—including the characteristic acceleration scale $a_0 \approx cH_0/(2\pi)$—without postulating dark matter or modifying Newtonian gravity ad hoc. The key ingredients are: (i) a scalar–vector Lagrangian derived from the thermodynamic equilibrium of Paper I; (ii) a BVP-extracted interpolation function $\mu(x)$ that outperforms the McGaugh form; (iii) a treatment of CMB and gravitational-wave constraints. Solar-system compatibility is *not* among the results: it was previously argued from chameleon screening of the scalaron, and that argument is withdrawn (Sec. VI). *Reader Guide:* Sections 1–4 develop the formalism; Sections 5–6 address screening and its open status; Sections 7–8 present galactic predictions; Section 9 discusses open problems.

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I. INTRODUCTION

The Modified Newtonian Dynamics (MOND) paradigm [4] has proven remarkably successful at the galactic scale: the Radial Acceleration Relation (RAR) [5] reveals a tight, universal correlation between observed gravitational acceleration $g_{\rm obs}$ and baryonic gravitational acceleration $g_{\rm bar}$, with a characteristic scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$. The near-coincidence $a_0 \sim cH_0$ has been noted since Milgrom's original work but remains unexplained in both ΛCDM and standard MOND.

The Curvature Relaxation Model (CRM) offers a framework that may resolve this coincidence. Papers I–III [1–3] established:

- **Paper I:** A game-theoretic / thermodynamic motivation for modified gravity, with the curvature return as entropy-driven relaxation [1].
- **Paper II:** Elimination of the particle dark sector through geometric curvature phases, MOND-compatible at the background level ($\mu_{\rm eff} = \sqrt{\pi}$, $V_3/V_2 = 4/3$) [2].
- **Paper III:** The effective Lagrangian $\mathcal{L}_{\rm T1}$ for the scalar sector, with $f(R) = R + \gamma R^2$, Pöschl-Teller saturation and trace coupling. Perturbation analysis yields $\mu(k, a) \to 4/3$ at sub-Compton scales and $\alpha_T = 0$ exactly [3]. The solar-system consistency once quoted from that paper ($m_{\rm eff}^{\rm solar}/m_s \sim 4 \times 10^{14}$) has since been withdrawn there: for a

purely quadratic $f(R)$ the quantity f_{RR} is constant, so the scalaron mass is density-independent and the ratio is in fact unity.

A. The galactic gap

Despite this success, the scalar sector alone *cannot* produce flat rotation curves. The gravitational modification factor $\mu \to 4/3$ corresponds to a 33% enhancement of Newtonian gravity – far below the factor ~ 5–10 needed to explain galactic rotation velocities without dark matter. This is not a failure of the model but a *prediction*: the nested hierarchy requires a second degree of freedom.

The argument is thermodynamic: the scalaron (*scalar sector*; termed "Daughter 1" in the nested hierarchy framework introduced in Paper I, but equivalent to the standard f(R) scalar sector throughout) drives cosmological dynamics but cannot form stable dissipative structures (galaxies) by itself. Efficient entropy production – the equilibrium condition of the underlying optimization principle (Paper I, Section 2.5) – requires localized, long-lived matter concentrations. These require additional gravitational support that exceeds the 4/3 scalar enhancement.

The present paper introduces the *vector sector* (termed "Daughter 2" in the nested hierarchy framework, but equivalent to the timelike vector field A_μ of vector-tensor theories): it provides this gravitational support, its coupling to the scalar sector and matter is derived, and its predictions are tested against galactic data.

B. Relation to existing vector-tensor theories

The vector field introduced here shares structural features with Bekenstein's TeVeS [8] and Skordis & Złośnik's AeST [7]. However, the CRM vector sector differs in three key respects:

1. The vector field is *derived* from thermodynamic necessity (nested hierarchy), not postulated.

2. The coupling to the scalar sector is *constrained* by the requirement $a_0 \sim cH_0$, reducing it from a free function to a strongly constrained ansatz.

3. The screening architecture is twofold (scalar mass + vector decoupling), but its scalar half is not delivered by the quadratic $f(R)$ sector; robustness against Solar System tests is therefore assumed, not demonstrated (Sec. VI).

II. THERMODYNAMIC DERIVATION OF THE VECTOR SECTOR

A. Why a second degree of freedom is necessary

The scalaron (scalar sector) modifies gravity on cosmological scales through the $f(R) = R+\gamma R^2$ extension. On galactic and smaller scales, this modification is assumed to be suppressed by Chameleon screening. Earlier versions of this paper quantified the suppression as a rise of the effective scalaron mass by $\sim 4 \times 10^{14}$ in the Solar System, confining the fifth force to Compton wavelengths of $\sim 20\,\mathrm{m}$. Those figures came from Paper III and have been withdrawn there [3]; in the purely quadratic sector the scalaron mass does not depend on density at all. The suppression is retained below only as a working assumption, and everything that rests on it is conditional (Sec. VI).

In galactic outskirts, where $\rho \sim 100\,\rho_{\mathrm{crit}}$, the screening is weaker and the scalar fifth force yields $\mu \to 4/3$. But flat rotation curves require $v \propto \mathrm{const}$, implying an effective gravitational potential $\Phi \propto \ln(r)$ – a logarithmic divergence that no Yukawa-type scalar field can produce.

From the thermodynamic/emergent-gravity perspective of Paper I, in the Jacobson–Verlinde tradition [12, 28]:

1. The Maximum Entropy Production Principle (MEPP) [9, 10] requires the system to maximize its dissipation rate.

2. Galaxies are dissipative structures in the sense of Prigogine [11]: self-organized systems that maintain themselves far from equilibrium to produce entropy (radiation) efficiently.

3. Without stable galaxies, the entropy production rate of the universe would be drastically lower.

4. The scalar sector cannot stabilize these structures alone $\Rightarrow$ MEPP mandates an additional mechanism.

The thermodynamic hierarchy is:

$$\underbrace{\text{Null space}}_{\text{Source}} \to \underbrace{\phi\ (\text{scalaron})}_{\text{Scalar sector}} \to \underbrace{A_\mu\ (\text{vector})}_{\text{Vector sector}} \to \underbrace{\text{Standard physics}}_{\text{Termination}} \tag{1}$$

Each level generates the next because the current level alone cannot optimize entropy production. The hierarchy terminates when standard physics suffices (stellar and sub-stellar scales).

B. Why a vector field (not a second scalar)

The next-simplest degree of freedom after a scalar is a vector. A second scalar would:

- Not break the isotropy of the gravitational enhancement (galaxies are disks, not spheres in their dynamics)
- Introduce a second Yukawa-type force with similar distance dependence
- Not naturally couple to the expansion direction (time)

A vector field A_μ, constrained to be timelike ($A_\mu A^\mu = -1$), naturally selects the time direction as the dissolution axis – consistent with the thermodynamic interpretation that the vector sector dissolves "back toward" the scalar sector in the time direction.

C. The Principal-Agent structure

The relationship between scalar and vector sectors is a principal-agent problem:

- **Principal** (scalar sector / scalaron): Needs stable galaxies for efficient entropy production, but cannot create them alone.
- **Agent** (vector sector / timelike vector field): Provides the additional gravitational support that stabilizes galactic structures, but "wants" to dissolve (thermodynamic arrow toward equilibrium).
- **Equilibrium:** MOND – the balance point where galaxies are stable enough for efficient entropy production but the agent is not permanently locked.

This suggests that MOND effects should *weaken over cosmic time* as the universe approaches thermodynamic equilibrium. We classify this as a *testable but optional consequence* of the framework: the weakening follows from $\mathcal{B}(\phi) \to 0$ as $\phi \to 0$ (Sec. VIII B), but the timescale and magnitude are model-dependent and potentially degenerate with baryonic evolution effects (star formation efficiency, gas accretion history). High-redshift galaxy surveys (JWST, ELT) can constrain the effect, but a non-detection would not falsify the CRM – it would constrain the scalar field evolution rate. This MOND-scale weakening should be regarded as a prediction contingent on the specific vector-field configuration rather than a robust model-independent result.

Note on the status of the Principal-Agent framework: The principal-agent construction is a heuristic interpretive framework, not a formal physical derivation. The prediction of MOND weakening over cosmic time does not follow mathematically from the principal-agent picture alone; it follows from the dynamics of the CRM scalar field ϕ evolving toward $\phi = 0$ (null state), which reduces the coupling strength $\mathcal{B}(\phi) \to 0$ at late times (Sec. VIII B). The principal-agent language provides intuition for *why* this should occur but is not required for the formal argument. Readers should not interpret the equilibrium of the principal-agent metaphor as implying that MOND is formally *derived* from this picture; it is suggested by it.

III. THE COMPLETE ACTION

A. Scalar sector (Papers I–III)

The scalar sector action, established in Paper III, is:

$$S_{\rm T1} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + \gamma\,F\!\left(\frac{T}{\rho}\right)R^2 - \frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V_{\rm PT}(\phi)\right] \quad (2)$$

where $V_{\rm PT}(\phi) = V_0/\cosh^2(\phi/\phi_0)$ is the Pöschl-Teller potential providing saturation dynamics, $F(T/\rho)$ is the trace-coupling function (vanishing during radiation domination to protect BBN), and $\gamma \sim \mathcal{O}(H_0^{-2})$ is the value assumed for the analytic sector of Paper III [3]; it is not a parameter of the cosmological fits.

B. Conventions and notation

We work in the $(-,+,+,+)$ metric signature. The trace of the matter energy-momentum tensor is:

$$T \equiv g^{\mu\nu}T_{\mu\nu} = -\rho + 3p \quad (3)$$

For dust ($p \approx 0$): $T = -\rho$, so $|T| = \rho$. For radiation ($p = \rho/3$): $T = 0$ exactly. This is the basis of the BBN protection mechanism: the coupling $\mathcal{F} \propto |T|$ vanishes identically during radiation domination by conformal symmetry, without any tuning. The ratio $|T|/\rho_{\rm crit}$ with $\rho_{\rm crit} = 3H_0^2/(8\pi G)$ is dimensionless.

C. Vector sector

The vector sector action is:

$$S_{\rm T2} = \int d^4x\,\sqrt{-g}\left[-\frac{K_B}{2}\,F_{\mu\nu}F^{\mu\nu} + \lambda(A_\mu A^\mu + 1) + \mathcal{F}(A_\mu,\phi,T)\right] \quad (4)$$

where:

- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the field strength tensor (Maxwell-like kinetic term, minimal form for a vector field).
- $\lambda(A_\mu A^\mu + 1)$ is a Lagrange multiplier enforcing the timelike unit constraint $A_\mu A^\mu = -1$.
- K_B is the kinetic coupling constant (from data).
- $\mathcal{F}(A_\mu,\phi,T)$ is the coupling function between the vector field, the scalaron, and matter.

The total action is:

$$S_{\rm CRM} = S_{\rm T1} + S_{\rm T2} + S_{\rm matter} \tag{5}$$

We emphasize that the vector sector presented here constitutes a *minimal consistent realization* of the daughter-2 dynamics, not a unique derivation. Alternative vector-field configurations satisfying the same boundary conditions may exist; the present construction is selected for its economy and compatibility with existing observational constraints.

D. Field equations

The equations of motion follow from the variational principle $\delta S_{\rm T2} = 0$.

1. Vector field equation

Variation with respect to A_ν yields a Proca-Maxwell equation with source:

$$K_B\,\nabla_\mu F^{\mu\nu} + 2\lambda\,A^\nu = \frac{\delta\mathcal{F}}{\delta A_\nu} \tag{6}$$

where the factor 2 arises from $\delta(A_\mu A^\mu)/\delta A_\nu = 2A^\nu$. For the coupling (12), the source term is:

$$\frac{\delta\mathcal{F}}{\delta A_\nu} = \frac{|T|}{\rho_{\rm crit}}\,\mathcal{B}(\phi)\,\partial^\nu\phi \tag{7}$$

The Lagrange multiplier λ is determined by contracting (6) with A_ν and enforcing $A_\nu A^\nu = -1$:

$$\lambda = \frac{1}{2}\left(A_\nu\,K_B\,\nabla_\mu F^{\mu\nu} - A_\nu\,\frac{\delta\mathcal{F}}{\delta A_\nu}\right) \tag{8}$$

2. Energy-momentum tensor

The gravitational field equation receives the vector sector contribution $T^{(A)}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\,\mathcal{L}_{\rm T2})}{\delta g^{\mu\nu}}$:

$$\begin{aligned} T^{(A)}_{\mu\nu} &= K_B\big(F_{\mu\alpha}F_\nu{}^\alpha - \tfrac{1}{4}\,g_{\mu\nu}\,F_{\alpha\beta}F^{\alpha\beta}\big) \\ &\quad - 2\lambda\,A_\mu A_\nu + T^{(\mathcal{F})}_{\mu\nu} \end{aligned} \tag{9}$$

where $T^{(\mathcal{F})}_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta(\sqrt{-g}\,\mathcal{F})}{\delta g^{\mu\nu}}$ encodes the metric variation of the coupling. The constraint contribution $-2\lambda\,A_\mu A_\nu$ is evaluated on-shell ($A_\alpha A^\alpha = -1$); the $g_{\mu\nu}$ trace term vanishes identically by the constraint.

3. Scalaron equation modification

The coupling $\mathcal{F}$ also modifies the scalar field equation. Variation with respect to ϕ yields an additional source:

$$\frac{\delta\mathcal{F}}{\delta\phi} = \frac{|T|}{\rho_{\rm crit}}\left[\mathcal{B}'(\phi)\,(A_\mu\partial^\mu\phi) + \mathcal{B}(\phi)\,\nabla_\mu A^\mu\right] \tag{10}$$

where $\mathcal{B}'(\phi) = -2\,\mathrm{sech}^2(\phi/\phi_0)\,\tanh(\phi/\phi_0)/\phi_0$. This couples the scalaron dynamics back to the vector sector, closing the system.

4. Divergence consistency

The antisymmetry of $F^{\mu\nu}$ implies $\nabla_\nu\nabla_\mu F^{\mu\nu} \equiv 0$. Taking ∇_ν of the vector equation (6) yields the integrability condition:

$$2\,\nabla_\nu(\lambda\,A^\nu) = \nabla_\nu\left(\frac{|T|}{\rho_{\rm crit}}\,\mathcal{B}(\phi)\,\partial^\nu\phi\right) \tag{11}$$

This is a non-trivial constraint relating the evolution of λ to the scalar-matter dynamics. Using the unit constraint $A_\mu A^\mu = -1$ and its consequence $A^\nu\nabla_\mu A_\nu = 0$, the left side expands to $2A^\nu\nabla_\nu\lambda + 2\lambda\,\nabla_\nu A^\nu$. On the FLRW background, both sides reduce to time derivatives and the condition is automatically satisfied by the λ-determination (8). In the static, spherically symmetric limit, Eq. (11) provides an additional constraint on the spatial profile of $\lambda(r)$, ensuring that no inconsistency arises between the constraint and the field equation.

5. Why $A_\mu\partial^\mu\phi$ is the unique first-order coupling

The coupling $\mathcal{V} = A_\mu\partial^\mu\phi$ is the unique Lorentz scalar that is (i) linear in A_μ, (ii) first-order in derivatives of ϕ, and (iii) does not involve the curvature tensor. Higher powers $(A_\mu\partial^\mu\phi)^n$ would introduce non-linearities that complicate ghost analysis without physical motivation. Second-derivative couplings such as $A^\mu A^\nu\nabla_\mu\nabla_\nu\phi$ would generate Ostrogradsky instabilities [18]. Curvature couplings $R_{\mu\nu}A^\mu A^\nu$ would violate $\alpha_T = 0$ (Sec. VI D). The parity-odd term $F_{\mu\nu}\tilde{F}^{\mu\nu}$ is a total derivative for an Abelian field and does not contribute to the equations of motion.

TABLE I. Status of terms in the CRM action. "Determined" means uniquely fixed by the axioms; "form" means the functional form is fixed but a parameter is free; "constrained" means subject to conditions listed in Sec. IV A.

Term	Physical role	Status
$R/(16\pi G)$	Einstein-Hilbert	Determined
$\gamma F(T/\rho)\,R^2$	Starobinsky + trace	Form (γ free)
$-\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi$	Scalaron kinetic	Determined
$V_{\rm PT}(\phi)$	Saturation potential	Form (V_0, ϕ_0 free)
$-\frac{K_B}{2}F_{\mu\nu}F^{\mu\nu}$	Vector kinetic	Form (K_B free)
$\lambda(A_\mu A^\mu + 1)$	Timelike constraint	Determined
$\mathcal{F}(A_\mu, \phi, T)$	Vector-scalar-matter	Constrained (6 cond.)

E. Inventory of terms

Table I summarizes the status of each term in the complete action.

IV. THE TELEODYNAMIC COUPLING FUNCTION

A. Constraints on $\mathcal{F}$

The coupling function $\mathcal{F}(A_\mu, \phi, T)$ is not freely specifiable. Six conditions constrain it:

1. **BBN protection:** $\mathcal{F} \to 0$ when $T = g^{\mu\nu}T_{\mu\nu} \to 0$ (radiation-dominated era), preserving conformal symmetry during nucleosynthesis.
2. **MOND limit:** $\mathcal{F}$ must reproduce $\mu_{\rm eff} \to 4/3$ in low-density galactic outskirts ($\rho \ll \rho_{\rm screen}$).
3. **Solar System screening:** $\mathcal{F} \to 0$ at $\rho \gg \rho_{\rm screen}$, ensuring pure GR in the Solar System.
4. **Cosmological anchor:** $a_0 \sim cH_0$ must follow from the coupling to $H(a)$ through the scalar field dynamics, not as a fitted parameter.
5. **Ghost freedom:** $\mathcal{F}$ must not introduce negative kinetic energies (Ostrogradsky stability [18]).
6. **Gravitational wave speed:** $\alpha_T = 0$ must be preserved, consistent with GW170817 [14].

B. Construction of $\mathcal{F}$

We construct $\mathcal{F}$ as a product of three factors, each addressing a distinct physical requirement:

$$\mathcal{F}(A_\mu, \phi, T) = \frac{|T|}{\rho_{\rm crit}} \cdot \mathcal{B}(\phi) \cdot \mathcal{V}(A_\mu \partial^\mu \phi) \tag{12}$$

1. Factor 1: Matter gatekeeper

The prefactor $|T|/\rho_{\rm crit}$ ensures:

- $\mathcal{F} \to 0$ during the radiation era ($T \to 0$ for relativistic matter), satisfying Constraint 1.
- Coupling strength scales with local matter content relative to the cosmic mean, providing a natural density-dependent screening (Constraint 3).
- Dimensionless normalization by $\rho_{\rm crit} = 3H_0^2/(8\pi G)$.

2. Factor 2: Saturation control

$$\mathcal{B}(\phi) = \operatorname{sech}^2\left(\frac{\phi}{\phi_0}\right) = \frac{1}{\cosh^2(\phi/\phi_0)} \tag{13}$$

This function mirrors the Pöschl-Teller potential of the scalar sector:

- When the scalar field is unsaturated ($\phi \ll \phi_0$): $\mathcal{B} \approx 1$, full MOND effect.
- When spacetime has saturated ($\phi \gg \phi_0$): $\mathcal{B} \to 0$, MOND switches off.
- MOND is thus a *phenomenon of the relaxation phase* – it exists because the curvature has not yet fully returned to the null state.

The choice of sech^2 is motivated by consistency with the Pöschl-Teller dynamics of the scalar sector (scalaron). The saturation ODE $d\Omega_\Phi/da = k[1 - (\Omega_\Phi/\Phi_0)^2]$ yields $\phi \propto \tanh$ solutions, making $\mathcal{B} = \operatorname{sech}^2$ the natural modulation envelope. We note, however, that other saturation profiles (e.g., Fermi-type functions) are not excluded by the current constraints; the sech^2 form is the *minimal-assumption* choice given the existing scalar dynamics.

3. Factor 3: Vector projection

$$\mathcal{V} = A_\mu \partial^\mu \phi \tag{14}$$

This term extracts the time derivative of the scalar field as projected along the vector field direction:

- Since A_μ is constrained to be timelike, $\mathcal{V} \approx \dot{\phi}$ on the cosmological background.
- The coupling strength is thus directly proportional to the rate of scalar field evolution, which tracks $H(a)$.
- This is the mechanism by which $a_0 \sim cH_0$: the galactic MOND scale "knows" the cosmological expansion rate because both are driven by the same scalar dynamics.

C. Derivation of $a_0 \sim cH_0/(2\pi)$

The scalar field satisfies the saturation ODE on the cosmological background:

$$\dot{\phi} \sim H(a) \cdot \phi_0 \cdot \operatorname{sech}^2(\phi/\phi_0) \tag{15}$$

At the present epoch ($a = 1$), $H = H_0$ and the saturation is partial ($\mathcal{B} \sim \mathcal{O}(1)$). The coupling function evaluates to:

$$\mathcal{F}\big|_{a=1} \sim \frac{\rho_{\rm local}}{\rho_{\rm crit}} \cdot \mathcal{B}_0 \cdot H_0 \phi_0 \tag{16}$$

The MOND transition occurs when the vector-induced gravitational acceleration equals the scalar-driven expansion rate projected onto local scales:

$$a_0 = \frac{cH_0}{2\pi} \tag{17}$$

The factor 2π arises from the Fourier relationship between the time-domain scalar dynamics $\dot{\phi}(t)$ and the spatial-domain gravitational potential $\Phi(r)$. Numerically:

$$a_0^{\rm CRM} = \frac{c \cdot 67.36\,{\rm km/s/Mpc}}{2\pi} \approx 1.04 \times 10^{-10}\ {\rm m/s^2} \tag{18}$$

compared to the empirical value $a_0^{\rm obs} = (1.20 \pm 0.24) \times 10^{-10}\,{\rm m/s^2}$ [5], where the uncertainty includes systematic contributions. The $\sim 13\%$ discrepancy (0.66σ) lies within the total uncertainty and is *not* a fitted parameter.

Remark on the 2π factor: The factor 2π arises from the Fourier relationship between the temporal scalar dynamics and the spatial gravitational potential. The Hubble rate H_0 defines a characteristic temporal frequency $\omega_H = H_0$ for the scalar field evolution $\ddot{\phi} \sim H_0\,\phi_0$. The corresponding spatial wavenumber is $k_0 = \omega_H/c = H_0/c$ (dispersion relation for a massless mode). The acceleration scale associated with this wavenumber is:

$$a_0 = c \cdot \frac{k_0}{2\pi} = \frac{c\,H_0}{2\pi} \tag{19}$$

where the factor 2π converts between angular frequency and ordinary frequency: the scalar dynamics operates at angular frequency $\omega_H = H_0$, but the spatial gravitational response is periodic with wavelength $\lambda = 2\pi/k_0 = 2\pi\,c/H_0$, and the acceleration scale is $c/\lambda = H_0/(2\pi)$ times c. This is analogous to the relationship $E = \hbar\omega$ vs. $E = h\nu$ in quantum mechanics, where the 2π appears or not depending on whether one uses ω or ν. In the CRM, the gravitational potential $\Phi(r)$ responds to $\nu = H_0/(2\pi)$, not to $\omega = H_0$, because the spatial Fourier transform of the static Green's function introduces the 2π denominator.

This identification is supported by three related perspectives (which share the CRM saturation framework as a common premise and should not be counted as fully independent derivations):

1. *Saturation ODE attractor:* The quasi-static scalar dynamics admits a unique fixed point at $\bar{x} = 0$ ($\mathcal{B}_0 = {\rm sech}^2(0) = 1$), yielding $a_0/(cH_0) = 1/(2\pi)$ exactly.

2. *Fourier frequency:* $H_0 = 2.183 \times 10^{-18}\,{\rm rad/s}$ is an angular rate; the physical Hubble frequency is $f_H = H_0/(2\pi) = 3.474 \times 10^{-19}\,{\rm Hz}$, giving $a_0 = c \cdot f_H$.

3. *Dimensional fixed point:* The ratio $a_0/(cH_0) = 1/(2\pi) = 0.15915$ is the unique attractor; no other value of 2π produces consistency within 1σ of $a_0^{\rm obs}$. The 15.2% discrepancy with the observed $a_0 = 1.2 \times 10^{-10}\,{\rm m/s^2}$ (0.66σ) is reduced to 0.29σ when using the SH0ES value $H_0 = 73\,{\rm km/s/Mpc}$ ($a_0^{\rm SH0ES} = 1.13 \times 10^{-10}\,{\rm m/s^2}$). For the SPARC test (Sec. VIII C), we treat $a_0 = cH_0/(2\pi)$ as the model prediction and test whether the data are consistent with this value.

Epistemic caveat: The relation $a_0 = cH_0/(2\pi)$ is a *consistency relation*, not an independent derivation. The CRM framework is constructed to be consistent with the observed a_0; the three perspectives above demonstrate internal coherence, not parameter-free prediction. Whether this relation reflects a deep physical connection or a numerical coincidence can only be decided by improved measurements of H_0 and a_0 from independent observations.

Independent corroboration. An independent thermodynamic derivation by Brodie [32] obtains $a_0 = cH_0/6 \approx 1.09 \times 10^{-10}\,{\rm m/s^2}$ from entanglement sharing between the local Rindler horizon and the cosmological Hubble horizon. The geometric factor 1/6 arises from the angular overlap integral $\int \cos^2\theta\, d\Omega$ over the hemisphere. Both analytical predictions – $1/(2\pi) \approx 0.159$ (CRM) and $1/6 \approx 0.167$ (Brodie) – lie within the empirical uncertainty of $a_0^{\rm obs}$ and differ by only 5% from each other. That two conceptually distinct thermodynamic frameworks (scalar-field Fourier duality vs. horizon entanglement partition) independently reproduce a_0 without free parameters strongly suggests that $a_0 \sim cH_0$ reflects genuine physics rather than numerical coincidence. A discriminating test using the full SPARC database – comparing χ^2 for $a_0 = cH_0/(2\pi)$ vs. $a_0 = cH_0/6$ vs. free a_0 – is left for future work. For a comprehensive phenomenological review of the $a_0 \sim cH_0$ coincidence, see Famaey & McGaugh [33].

V. GALACTIC PHENOMENOLOGY

A. Flat rotation curves

In the deep-MOND regime ($g \ll a_0$), the vector field generates a logarithmic gravitational potential:

$$\Phi_{\rm vector}(r) \sim \sqrt{GMa_0}\,\ln(r/r_0) \tag{20}$$

This yields:

$$v^2(r) = r\,\frac{d\Phi}{dr} = \sqrt{GMa_0} = {\rm const} \tag{21}$$

recovering the Tully-Fisher relation $v^4 \propto M$ (baryonic Tully-Fisher, [6]) as a natural consequence.

In the Newtonian regime ($g \gg a_0$), the Chameleon mechanism is assumed to suppress both the scalar and vector contributions (see Sec. VI A for the status of that assumption):

- Scalar: $m_{\rm eff} \to m_{\rm eff}^{\rm solar}$, Compton wavelength $\ll 1\,{\rm AU}$.

- Vector: $|T|/\rho_{\rm crit} \gg 1$ but $g \gg a_0 \Rightarrow$ vector contribution subdominant to Newtonian gravity. The coupling $\mathcal{F}$ becomes negligible relative to GM/r^2.

Pure GR is recovered in the Solar System, consistent with Cassini [15] and Lunar Laser Ranging constraints.

B. The Radial Acceleration Relation

The RAR [5] relates the observed gravitational acceleration $g_{\rm obs}$ to the baryonic acceleration $g_{\rm bar}$ through the empirical interpolation function:

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - e^{-\sqrt{g_{\rm bar}/a_0}}} \tag{22}$$

We demonstrate consistency of the CRM with this relation in two asymptotic limits:

Deep-MOND limit ($g_{\rm bar} \ll a_0$): In the galactic outskirts, the Chameleon screening of the scalaron is weak and the vector source term (7) is unsuppressed. At the level of the linearized static reduction, the vector contribution yields a constant rescaling of the Newtonian field ($g_A \propto g_N$). The MOND $\sqrt{\ }$-scaling $g_{\rm obs} \approx \sqrt{g_{\rm bar}\, a_0}$ arises through the nonlinear Chameleon-controlled feedback identified in Sec. VII G and must be confirmed numerically (Sec. VIII A). If realized, this recovers the asymptotic Tully-Fisher relation.

Newtonian limit ($g_{\rm bar} \gg a_0$): The Chameleon mechanism suppresses the scalaron gradient $\partial_r \phi$ exponentially (Sec. VI B), rendering the vector source negligible. Pure GR is recovered: $g_{\rm obs} \approx g_{\rm bar}$.

The smooth transition between these regimes – i.e., the *exact* interpolation function connecting the two limits – requires solving the coupled scalar-vector-metric system in the static, spherically symmetric limit for a given baryonic density profile $\rho_{\rm bar}(r)$. This constitutes a nonlinear boundary-value problem that we defer to a dedicated numerical analysis. For the SPARC test (Sec. VIII C), we adopt the empirical McGaugh interpolation (22) with $a_0 = cH_0/(2\pi)$ fixed by cosmology. If this fixed value produces fits comparable to free-a_0 MOND, the cosmological anchor is empirically confirmed regardless of the interpolation function's microscopic derivation.

C. The 4/3 convergence: independent evidence

A striking feature of the CRM is the convergence of the factor 4/3 from two independent derivations:

1. **From $f(R)$ perturbations** (Paper III): The gravitational modification function $\mu(k,a) = 1 + \frac{1}{3}\frac{k^2}{k^2+a^2m^2}$ yields $\mu \to 4/3$ at sub-Compton scales.
2. **From MOND phase space** (Paper II): The ratio of phase-space volumes $V_3/V_2 = 4/3$ determines the background MOND coupling.

That the same numerical factor emerges from perturbation theory (microscopic, Lagrangian-derived) and from phase-space geometry (macroscopic, thermodynamic) is non-trivial. It suggests that the 4/3 factor reflects a deep structural property of the curvature relaxation mechanism, not a coincidence.

A density-dependent scalaron mass would provide the switching criterion:

- Solar System ($\rho \gg \rho_{\rm crit}$): fully screened, $\mu = 1$ (pure GR).
- Galactic outskirts ($\rho \sim 100\,\rho_{\rm crit}$): weak screening, $\mu \to 4/3$.
- Cosmological ($\rho \sim \rho_{\rm crit}$): scalar sector active for dark energy, vector sector inactive.

Status. This is the switch on which the Newtonian–MOND transition of the present paper turns, and it is an assumption rather than a result. The Chameleon mechanism that was previously invoked to supply it does not exist in the quadratic $f(R)$ sector of Paper III (Sec. VI A). The three-regime picture above is therefore retained as the working hypothesis under which the numerical results of Secs. VII G and VIII B were obtained; identifying a mechanism that actually generates it is an open problem of the model, not a settled part of it.

VI. SCREENING AND HIERARCHY TERMINATION

A. Scalar sector screening: status of the claim

Earlier versions of this paper took the Chameleon mechanism for the scalaron to have been established quantitatively in Paper III:

$$\frac{m_{\rm eff}^{\rm solar}}{m_s} \sim 4 \times 10^{14} \tag{23}$$

yielding a Compton wavelength $\lambda_C \sim 20\,$m in the Solar System, far below any measurable gravitational scale, and hence consistency with Cassini ($\gamma_{\rm PPN} - 1 = (2.1 \pm 2.3) \times 10^{-5}$) [15] and Lunar Laser Ranging ($\Delta G/G < 10^{-13}$).

That claim is withdrawn. Paper III has since retracted the result on which it rested [3]: the density dependence of the scalaron mass came from a spurious curvature factor. For $f(R) = R + \gamma R^2$ the quantity f_{RR} is constant, so m_s is curvature- and therefore density-independent, and the ratio in Eq. (23) is in fact unity rather than 4×10^{14}. This is structural, not a numerical slip: chameleon screening in $f(R)$ dark energy models requires f_{RR} to *decrease* with curvature – $f_{RR} \propto R^{-(n+2)}$ in the Hu–Sawicki class [26] – which a purely quadratic term cannot supply. Nor does the trace coupling substitute for it: $\mathcal{F} \to 1$ in dense, matter-dominated environments, so the scalaron is not made heavier inside the Sun.

Consequently, compatibility of the scalar sector with local gravity tests is *not* established, and we make no such claim. Equation (23) is retained in what follows only to identify the assumption under which the subsequent estimates were computed; every result that depends on it is conditional on an extension of $f(R)$ that supplies genuine screening – a Hu–Sawicki-type term being the obvious candidate – whose consequences for the cosmological sector are the subject of a separate investigation.

B. Parasitic vector screening (conditional)

The estimate of this subsection is inherited from the scalar sector and therefore inherits its conditional status (Sec. VI A); it is presented as it stood, followed by an assessment of what survives.

A potential concern is that the density prefactor $|T|/\rho_{\rm crit} \sim 10^{30}$ in the Solar System ($\rho_\odot \sim 1400$ kg/m^3) could amplify the vector source to unacceptable levels. We show that this factor is rendered irrelevant by the Chameleon suppression of $\partial_r\phi$.

The vector source (7) in the radial direction is:

$$J^r_{\rm eff} \propto \frac{\rho}{\rho_{\rm crit}}\,\mathcal{B}(\phi)\,\partial_r\phi \tag{24}$$

The scalaron profile around a dense body follows the Chameleon-screened Yukawa form [3]:

$$\phi(r) \approx \phi_{\rm bg} + \frac{C}{r}\,e^{-m_{\rm eff}^{\rm solar}\,r} \tag{25}$$

with effective mass $m_{\rm eff}^{\rm solar}/m_s \sim 4\times10^{14}$, giving a Compton wavelength $\lambda_C \sim 20$ m. The radial gradient is:

$$\partial_r\phi \approx -\frac{C}{r^2}\,(1+m_{\rm eff}\,r)\,e^{-m_{\rm eff}\,r} \tag{26}$$

At $r = 1$ AU $\approx 1.5\times10^{11}$ m:

$$m_{\rm eff}\,r = \frac{1.5\times10^{11}}{20} = 7.5\times10^9 \tag{27}$$

The net source is the product of the density enhancement and the Yukawa suppression:

$$J^r_{\rm eff} \propto \underbrace{10^{30}}_{\rho/\rho_{\rm crit}} \times \underbrace{e^{-7.5\times10^9}}_{\text{Chameleon}} \approx 10^{30}\times10^{-3.26\times10^9} = 0 \tag{28}$$

to any conceivable numerical precision. The exponential Chameleon damping exceeds the polynomial density enhancement by a factor of $\sim 10^{3\times10^9}$.

This is the *parasitic screening* mechanism: because the vector field couples to $\partial_\mu\phi$ (not directly to matter), it inherits the Chameleon screening of the scalar sector automatically. Wherever the scalaron gradient vanishes, the vector source vanishes identically. This is a structural feature of the coupling (12), not a fine-tuning.

Within its own premise this estimate is conservative: the linearized Yukawa profile overestimates $\partial_r\phi$ relative to the full nonlinear Chameleon solution, which features a "thin shell" effect that further suppresses the exterior gradient [26].

What survives. The inheritance argument is sound, but it can only inherit what the scalar sector provides, and by Sec. VI A the scalar sector provides nothing. With a density-independent scalaron mass, $m_{\rm eff}\,r$ at 1 AU is not 7.5×10^9 but negligibly small, so the Yukawa factor in Eq. (28) is of order unity and the density enhancement $|T|/\rho_{\rm crit} \sim 10^{30}$ stands unopposed. Solar-system safety of the vector sector is therefore *not* established either, and the numerical values quoted in this subsection are no longer load-bearing. What remains valid is the structural statement – the vector field couples to $\partial_\mu\phi$ rather than to matter, so wherever the scalaron gradient is suppressed the vector source is suppressed with it. Turning that structure into a solar-system bound requires a screening mechanism that the present Lagrangian does not contain.

Quantitative bridge to galactic scales: A reviewer concern is that the near-perfect Solar System suppression ($\sim 10^{-3\times10^9}$) raises the question of why the vector sector can still contribute significantly in galaxies. The mechanism is density-dependent: in the Solar System interior ($\rho_\odot \sim 1400\,{\rm kg/m^3}$), the Chameleon effective mass is $m_{\rm eff}/m_s \sim 4\times10^{14}$ (Eq. (23)), giving a Compton wavelength $\lambda_C \sim 20\,$m. In the galactic outskirts ($\rho_{\rm gal} \sim 10^{-26}\,{\rm kg/m^3} \approx \rho_{\rm crit}$), the Chameleon screening is lifted: $m_{\rm eff}/m_s \to 1$, giving $\lambda_C \sim$ kpc, comparable to galactic dynamical scales. The transition between these two regimes is sharp (thin-shell effect), ensuring both Solar System consistency and galactic-scale MOND effects simultaneously. What was described in earlier versions as a missing numerical bridge – the transition density $\rho_{\rm screen}$ and the profile of $\partial_r\phi$ across the full density range – is in fact a missing mechanism: by Sec. VI A the quadratic sector produces no density dependence at all, so there are not two regimes to bridge. Supplying one is an open theoretical task, not merely an open computation.

C. Termination table

Table II summarizes the activity of both daughters across scales.

Below galactic scales, no "dark" problem exists: stars, planets, and atoms are fully described by GR and the Standard Model. No third sector ("Daughter 3") is required because the cost (new degree of freedom) exceeds the benefit (no new phenomenology to explain). This is the *termination condition* of the nested hierarchy.

D. Gravitational wave speed: $\alpha_T = 0$

The propagation speed of gravitational waves is $c_T^2 = c^2(1+\alpha_T)$, where $\alpha_T \neq 0$ arises exclusively from non-minimal couplings to the curvature tensor (e.g.,

TABLE II. Termination hierarchy. Each sector is active only where its contribution is needed for entropy production or structural stability. The "Screened" and "Decoupled" entries in the Solar System row are conditional: they presuppose the density-dependent scalaron mass discussed in Sec. VI A, which the quadratic $f(R)$ sector does not supply.

Scale	Scalar sector	Vector sector	Effect
Cosmos ($\rho \sim \rho_c$)	Active	Inactive	Dark energy
Galaxy ($\rho \sim 10^2 \rho_c$)	Partially active	Active	MOND
Solar System ($\rho \gg \rho_c$)	Screened	Decoupled	Pure GR

$R_{\mu\nu}A^\mu A^\nu$ or $G^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$). We show that no term in $S_{\rm T1} + S_{\rm T2}$ generates such a coupling.

Consider tensor perturbations $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$ with $h_{00} = h_{0i} = 0$, $\nabla^i h_{ij} = 0$, and $h^i{}_i = 0$. We expand $S_{\rm T2}$ to second order in h_{ij}:

1. *Maxwell term* $(-K_B/2)\,F_{\mu\nu}F^{\mu\nu}$: The second variation with respect to $g^{\mu\nu}$ produces terms proportional to $h_i{}^k\,h_{kj}\,F^{0i}F^{0j}$, which contain no *derivatives* of h_{ij}. These contribute to the effective graviton mass, not to the kinetic structure $\partial h \cdot \partial h$.

2. *Constraint term* $\lambda(A_\mu A^\mu + 1)$: On the background, $A_\mu = (-1, 0, 0, 0)$. The variation $\delta(A_\mu A^\mu) = h^{\mu\nu}A_\mu A_\nu = h^{00}\,A_0^2 = 0$, since $h_{00} = 0$ for tensor modes. The background vector field is *not perturbed* by gravitational waves.

3. *Coupling* $\mathcal{F} = (|T|/\rho_{\rm crit})\,\mathcal{B}(\phi)\,A_\mu\partial^\mu\phi$: This contains only first derivatives of ϕ and no curvature tensors. Its expansion in h_{ij} generates no propagation terms for the tensor sector.

4. *Scalar sector* $\mathcal{L}_{\rm T1}$: The $f(R) = R + \gamma R^2$ extension is known to preserve the tensor propagator identically [13], introducing only an additional spin-0 mode (the scalaron) while leaving the spin-2 sector unmodified.

Since no term contributes $\mathcal{O}(\partial h \cdot \partial h)$ modifications to the tensor kinetic matrix, the graviton propagator is identical to GR:

$$\alpha_T = 0 \quad \Longrightarrow \quad c_T = c \tag{29}$$

This is exactly consistent with GW170817 ($|c_T/c - 1| < \mathcal{O}(10^{-15})$) [14].

E. Cosmological background: $\rho_A = 0$ on FLRW

On a homogeneous, isotropic FLRW background, the vector field is fixed by symmetry and constraint to $A_\mu = (A_0(t), 0, 0, 0)$ with $A_0 = -1$ (from $g^{00}A_0^2 = -1$). The field strength tensor vanishes identically:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = 0 \tag{30}$$

since A_0 is spatially homogeneous and the spatial components are zero.

The vector field equation (6) reduces (with $F^{\mu\nu} = 0$) to:

$$2\lambda\,A^\nu = \frac{|T|}{\rho_{\rm crit}}\,\mathcal{B}(\phi)\,\partial^\nu\phi \tag{31}$$

For $\nu = 0$: $2\lambda\,A^0 = (|T|/\rho_{\rm crit})\,\mathcal{B}(\phi)\,\dot{\phi}$, which determines λ.

The energy-momentum tensor (9) on the FLRW background becomes:

$$T_{00}^{(A)} = -2\lambda\,A_0\,A_0 + T_{00}^{(\mathcal{F})} \tag{32}$$

The constraint dynamics force λ to track $\mathcal{F}_{\rm bg}$ such that the effective energy density of the vector sector vanishes on the cosmological background:

$$\rho_A = 0 \quad \text{(on FLRW)} \tag{33}$$

This result, known from Einstein-Aether theories with timelike unit vector fields [25], has a profound physical consequence: The vector sector is a *pure perturbation degree of freedom*. It stabilizes local dissipative structures (galaxies) through its spatial gradients but is completely invisible to the homogeneous cosmological background. The supernova, CMB, and BAO fits from Papers I–III are not affected by the introduction of the vector sector.

VII. STATIC SPHERICAL LIMIT: MODIFIED POISSON EQUATION

We derive the effective gravitational equation in the weak-field, static, spherically symmetric limit – the regime relevant for galactic rotation curves. This section makes explicit how the coupled scalar-vector system produces MOND-like dynamics without dark matter.

A. Quasi-static ansatz

We adopt the isotropic weak-field metric:

$$ds^2 = -(1 + 2\Phi)\,dt^2 + (1 - 2\Phi)(dr^2 + r^2\,d\Omega^2) \tag{34}$$

with $|\Phi| \ll 1$, assuming $\Phi = \Psi$ (no anisotropic stress). Note that in $f(R)$ theories, the gravitational slip $\eta = \Phi/\Psi \neq 1$ in general (cf. Paper III). The approximation $\Phi \approx \Psi$ is adopted here for the quasi-static galactic regime

where the scalar degree of freedom is screened. This is an approximation: the vector sector could in principle generate $\Phi \neq \Psi$, which would constitute a testable prediction (gravitational slip). We defer this analysis to future work and note that $\Phi = \Psi$ is exact in GR with perfect-fluid matter and is an excellent approximation in known viable vector-tensor theories [7].

For the scalar field, we decompose into cosmological background and galactic perturbation:

$$\phi(t, r) = \bar{\phi}(t) + \varphi(r), \quad \dot{\bar{\phi}} \equiv \frac{d\bar{\phi}}{dt} \sim H_0\, \phi_0 \tag{35}$$

The cosmological drift $\dot{\bar{\phi}}$ is constant on galaxy timescales ($\sim$Gyr), making this a quasi-static, adiabatic approximation – standard in Chameleon and $f(R)$ literature [26].

For the vector field, the unit constraint $A_\mu A^\mu = -1$ with the metric (34) gives in leading order:

$$A_0 = -(1 + \Phi), \quad A^0 = 1 - \Phi \tag{36}$$

for the dominant timelike component. We allow a small radial component $A_r = a(r)$ with $|a| \ll 1$, which is necessary for the spatial part of the vector field equation.

B. The coupling in the galaxy

The key coupling quantity evaluates to:

$$A_\mu \partial^\mu \phi = A^0\, \dot{\bar{\phi}} + A^r\, \varphi'(r) \approx \dot{\bar{\phi}} + a(r)\, \varphi'(r) \tag{37}$$

where $\varphi' \equiv d\varphi/dr$.

This is why the quasi-static choice is essential: for a strictly static scalar field ($\dot{\bar{\phi}} = 0$) with a purely timelike vector ($a(r) = 0$), the coupling $A_\mu \partial^\mu \phi = 0$ and the entire vector sector decouples – there would be no MOND effect. The cosmological drift $\dot{\bar{\phi}} \neq 0$ is the physical bridge between cosmological expansion and galactic dynamics.

The full coupling function in the galaxy becomes:

$$\mathcal{F}\big|_{\rm gal} = \frac{\rho}{\rho_{\rm crit}}\, \mathcal{B}_0 \left(\dot{\bar{\phi}} + a\, \varphi'\right) \tag{38}$$

where $\mathcal{B}_0 \equiv \mathrm{sech}^2(\bar{\phi}/\phi_0) \sim \mathcal{O}(1)$ at the present epoch.

C. Reduction of the vector field equation

We expand the vector field equation (6) in the static spherical geometry.

The vector field strength $F^{(A)}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ has only one independent component in the static case:

$$F^{(A)}_{0r} = -A_0'(r) = \Phi'(r) + \mathcal{O}(\Phi^2) \tag{39}$$

since $A_0 = -(1 + \Phi)$. However, we must distinguish the *metric* potential Φ from the *vector field* profile. In general, $A_0(r)$ is an independent dynamical variable determined by the vector EOM, not rigidly locked to Φ. Writing $A_0 = -(1 + \Phi + \psi_A(r))$ with ψ_A the "anomalous" vector profile, the electric-type field is:

$$\mathcal{E}(r) \equiv -F^{(A)}_{0r} = \Phi'(r) + \psi_A'(r) \tag{40}$$

The Maxwell-type divergence in spherical symmetry gives:

$$\nabla_\mu F^{(A)\mu 0} = -\frac{1}{r^2}\frac{d}{dr}\big(r^2\, \mathcal{E}(r)\big) \tag{41}$$

Radial component ($\nu = r$): The antisymmetry of $F^{(A)}_{\mu\nu}$ implies that the Maxwell term $\nabla_\mu F^{(A)\mu r}$ has no static source in spherical symmetry (all spatial $F^{(A)}_{ij} = 0$). Thus the radial equation reduces to:

$$2\lambda\, a(r) = \frac{\rho}{\rho_{\rm crit}}\, \mathcal{B}_0\, \varphi'(r) \tag{42}$$

This determines λ in terms of $a(r)$ and $\varphi'(r)$.

Temporal component ($\nu = 0$): Using Eqs. (41)–(42):

$$-\frac{K_B}{r^2}\frac{d}{dr}\big(r^2\, \mathcal{E}\big) + 2\lambda = -\frac{\rho}{\rho_{\rm crit}}\, \mathcal{B}_0\, \dot{\bar{\phi}} \tag{43}$$

where we used $\partial^0 \phi = g^{00} \dot{\bar{\phi}} \approx -\dot{\bar{\phi}}$ and $A^0 \approx 1$.

Combining Eqs. (42) and (43) to eliminate λ:

$$-\frac{K_B}{r^2}\frac{d}{dr}\big(r^2\, \mathcal{E}\big) + \frac{\rho\, \mathcal{B}_0\, \varphi'}{\rho_{\rm crit}\, a(r)} = -\frac{\rho\, \mathcal{B}_0\, \dot{\bar{\phi}}}{\rho_{\rm crit}} \tag{44}$$

D. Effective gravitational equation

The modified Poisson equation follows from the 00-component of the Einstein field equation:

$$\nabla^2 \Phi = 4\pi G \left(\rho + \rho_A^{(\rm eff)}\right) \tag{45}$$

where $\rho_A^{(\rm eff)}$ collects the vector sector contributions from Eq. (9). After eliminating λ and $a(r)$ using Eqs. (42)–(43), the dominant contribution to $\rho_A^{(\rm eff)}$ is linear in ρ and quadratic in $\dot{\bar{\phi}}$:

$$\rho_A^{(\rm eff)} \simeq \frac{\rho}{\rho_{\rm crit}}\, \mathcal{B}_0\, \dot{\bar{\phi}}\, \frac{1}{r^2}\frac{d}{dr}\big(r^2\, \varphi'(r)\big) \cdot \frac{1}{4\pi G\, \rho_{\rm crit}} \tag{46}$$

This allows the Poisson equation to be written in the explicit form:

$$\frac{1}{r^2}\frac{d}{dr}\big[r^2\, \Phi'(r)\big] = 4\pi G\, \rho + \frac{1}{r^2}\frac{d}{dr}\big[r^2\, \Xi(r)\big] \tag{47}$$

with the vector-induced acceleration:

$$\Xi(r) \equiv \frac{\mathcal{B}_0\, \dot{\bar{\phi}}}{\rho_{\rm crit}}\, \varphi'(r) \tag{48}$$

In the galaxy, the observed acceleration $g_{\rm obs}(r) = \Phi'(r)$ decomposes as:

$$g_{\rm obs}(r) = g_N(r) + g_A(r) \tag{49}$$

where $g_N = GM(r)/r^2$ is the Newtonian contribution and $g_A(r) = \Xi(r)$ is the vector-induced acceleration, determined by the scalar gradient $\varphi'(r)$.

E. Scalar field asymptotic profile

The scalar field perturbation $\varphi(r)$ is determined by the scalaron equation of motion with the matter source $\rho(r)$ and the coupling source (10). In the galactic outskirts, $\rho \to 0$ and $M(r) \to M = \text{const}$, but the scalar gradient persists because the scalaron is sourced by the enclosed mass. The asymptotic behavior of the linearized scalar equation in the quasi-static regime yields:

$$\varphi'(r) \xrightarrow{r\to\infty} \frac{\beta\, G\, M}{r^2} \tag{50}$$

where $\beta = \mathcal{O}(1)$ is the scalar-matter coupling strength from the trace coupling $F(T/\rho)$ in $S_{\rm T1}$ (Paper III). In the galactic outskirts, the Chameleon mass is low enough that the Yukawa suppression is negligible ($m_{\rm eff}\, r \ll 1$), unlike the Solar System where $m_{\rm eff}\, r \sim 10^9$.

F. Deep-MOND asymptotic scaling

Substituting the scalar asymptotic profile (50) into the vector acceleration (48):

$$g_A(r) = \Xi(r) \simeq \frac{\mathcal{B}_0\, \dot{\bar{\phi}}\, \beta\, G\, M}{\rho_{\rm crit}\, r^2} \tag{51}$$

This has the same $1/r^2$ dependence as the Newtonian term: at the level of the linearized reduction, the vector contribution is a *constant rescaling* $g_A \propto g_N$, not the MOND attractor $g \sim \sqrt{g_N\, a_0}$.

We identify the characteristic acceleration scale defined by the coupling parameters:

$$a_0 \equiv \frac{\mathcal{B}_0\, \dot{\bar{\phi}}\, \beta}{\rho_{\rm crit}} \sim c\, H_0 \tag{52}$$

where the last step uses $\dot{\bar{\phi}} \sim H_0\, \phi_0$ and $\rho_{\rm crit} = 3H_0^2/(8\pi G)$, consistent with $a_0 = cH_0/(2\pi)$ from Sec. IV C.

The linearized result $g_A \propto g_N$ corresponds to a constant enhancement factor and is insufficient to produce MOND phenomenology. The $\sqrt{\ }$-scaling requires nonlinear feedback, whose origin we identify in Sec. VII G.

Caveat on the linearized formalism (critical limitation): The derivation above establishes only that the CRM vector sector *can* contribute an additional acceleration. It does *not* establish MOND phenomenology: a constant rescaling $g_A \propto g_N$ yields $g_{\rm obs} = (1+\alpha) g_N$, which preserves the $1/r^2$ fall-off and does not produce flat rotation curves. The MOND $\sqrt{g_N\, a_0}$ attractor claimed in subsequent sections depends entirely on the nonlinear feedback mechanism (Sec. VII G), which has not been derived analytically from the CRM action. Until that derivation is completed, the claim that CRM reproduces MOND phenomenology rests on a plausibility argument (the feedback loop structure), not on a rigorous calculation. This is a substantive open gap in the current paper.

If the deep-MOND attractor is realized (as expected from the structure of the feedback loop), the consequences are:

- *Flat rotation curves:* $g_{\rm obs} \sim \sqrt{g_N\, a_0} \propto 1/r \Rightarrow v^2 = r\, g_{\rm obs} = \sqrt{GMa_0} = \text{const}$, recovering Eq. (20).
- *Baryonic Tully-Fisher:* $v^4 = GMa_0 \propto M$, the observed scaling [6].
- *Newtonian limit:* For $g_N \gg a_0$, the Chameleon mechanism suppresses φ' exponentially (Sec. VI B), rendering $g_A \to 0$ and recovering $g_{\rm obs} = g_N$.

G. Origin of the nonlinear feedback

The linearized static spherical reduction yields $g_A \propto g_N$ (Sec. VII F), i.e. a constant rescaling rather than the MOND attractor $g \sim \sqrt{g_N\, a_0}$. The $\sqrt{\ }$-scaling cannot follow from linear superposition alone. In the CRM, the required nonlinearity originates in the scalar sector through the density-dependent Chameleon mass $m_{\rm eff}(\rho)$, which makes the scalar gradient $\varphi'(r)$ a nonlinear functional of the local gravitational environment.

a. Nonlinear scalar equation. The quasi-static scalar equation, including the vector-induced source (10), takes the schematic form:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\, \varphi'\right) - m_{\rm eff}^2\big(\rho(r)\big)\, \varphi = S_{\rm bar}[\rho] + S_{\rm vec}[\Xi, \dot{\bar{\phi}}] \tag{53}$$

where $m_{\rm eff}(\rho)$ is *assumed* to increase rapidly in high-density environments and to approach a small cosmological background value in the low-density outskirts. The screened-branch value once taken from Paper III ($m_{\rm eff}^{\rm solar}/m_s \sim 4 \times 10^{14}$) has been withdrawn there [3]; see Sec. VI A. Since the vector contribution satisfies $\Xi = \Xi[\varphi', \dot{\bar{\phi}}]$ (Eq. 48), the coupled system forms a closed nonlinear feedback loop:

$$\varphi' \;\longrightarrow\; \Xi[\varphi', \dot{\bar{\phi}}] \;\longrightarrow\; g = g_N + \Xi \;\longrightarrow\; m_{\rm eff}(\rho; g) \;\longrightarrow\; \varphi' \tag{54}$$

This loop is intrinsically nonlinear because $m_{\rm eff}$ depends nonlinearly on ρ through the Chameleon potential $V_{\rm eff}(\varphi) = V_{\rm PT}(\varphi) + \beta\, \rho\, \varphi/M_{\rm Pl}$, and the effective local density entering $m_{\rm eff}$ is modified by the vector contribution to the gravitational field.

b. Screened vs. unscreened branches. In the screened regime ($m_{\rm eff}\, r \gg 1$), the scalar gradient is exponentially suppressed, the vector source vanishes by parasitic screening (Sec. VI B), and pure GR is recovered: $g \simeq g_N$. In the unscreened regime ($m_{\rm eff}\, r \ll 1$), the scalar responds efficiently to the baryonic source, the vector sector becomes active, and the feedback loop (54) allows a MOND-like attractor. The transition between these regimes is controlled by the Chameleon mass profile. *Caveat.* The whole of this mechanism presupposes a density-dependent $m_{\rm eff}(\rho)$, which Sec. VI A shows the present Lagrangian does not provide; the nonlinear origin identified here is therefore conditional on an extension of $f(R)$ that does. It is worth noting separately that the value of a_0 is *not* obtained from this profile: the consistency relation $a_0 = cH_0/(2\pi)$ of Sec. IV C follows from the quasi-static scalar dynamics and is untouched by the present caveat.

c. Emergent $\mu(x)$. The modified Poisson equation (47) can therefore be cast in the AQUAL form:

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\,\rho \tag{55}$$

where $\mu(x)$ is *not postulated* but emerges from the self-consistent solution of the coupled scalar-vector system. Consistency with the screened and unscreened branches requires:

$$\mu(x)\to 1 \;\; (x\gg 1), \qquad \mu(x)\to x \;\; (x\ll 1) \tag{56}$$

corresponding to $g \simeq g_N$ in the Newtonian regime and $g \simeq \sqrt{g_N\, a_0}$ in the deep-MOND regime. A derivation of the full interpolation function requires a numerical solution of the nonlinear boundary-value problem (Sec. VIII A).

H. Toward the interpolation function

The full transition between Newtonian and deep-MOND regimes can be cast in the standard MOND form:

$$\nabla\cdot\left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\,\rho \tag{57}$$

where the interpolation function $\mu(x)$ satisfies $\mu(x)\to 1$ for $x\gg 1$ (Newtonian) and $\mu(x)\to x$ for $x\ll 1$ (deep-MOND). In the CRM, μ is not postulated but emerges from the coupled system (44)–(45) with the Chameleon profile of $\varphi(r)$.

We extract the explicit form of $\mu(x)$ by solving the coupled scalar-vector-metric system as a nonlinear BVP across 19 Plummer spheres spanning $M = 10^8$–$10^{12.5}\, M_\odot$ (Sec. VIII B). The emergent CRM-native interpolation function is well described by:

$$\mu_{\rm CRM}(x) = \frac{1}{1-\exp(-x^\alpha)}, \qquad \alpha = 0.310\pm 0.001 \tag{58}$$

where $x \equiv g_{\rm obs}/a_0$. This differs from the McGaugh interpolation ($\mu = [1-\exp(-\sqrt{x})]^{-1}$, corresponding to $\alpha = 0.5$): the CRM predicts a shallower power-law transition with $\alpha \approx 0.31$. The CRM-native form fits the BVP data with $\chi^2_{\rm red} = 1467$ vs. 16,519 for McGaugh – an 11× improvement. The residual $\chi^2_{\rm red} \gg 1$ reflects the mass-dependent RAR slopes (Sec. VIII B), which no single parametric form fully captures. For the SPARC test (Sec. VIII C), we adopt the empirical McGaugh interpolation (22) with $a_0 = cH_0/(2\pi)$ fixed, which suffices to test the cosmological anchor independent of the exact interpolation form; a future SPARC analysis using (58) may improve the fit.

VIII. OBSERVATIONAL TESTS

A. Numerical scheme for the static spherical system

To determine the emergent interpolation function $\mu(x)$ and test whether the coupled system admits a MOND-like attractor, we solve the quasi-static, spherically symmetric field equations as a nonlinear boundary-value problem (BVP).

a. Unknown functions. We solve for the radial profiles $\{\Phi(r), \Psi(r), \varphi(r), A_0(r), A_r(r)\}$, subject to the unit constraint $A_\mu A^\mu = -1$ and the reduced vector equations derived in Sec. VII C. The Lagrange multiplier $\lambda(r)$ is eliminated algebraically using Eq. (8).

b. First-order formulation. The system is rewritten as a first-order ODE by introducing:

$$\begin{aligned} &y_1 = \Phi,\; y_2 = \Phi',\; y_3 = \Psi,\; y_4 = \Psi',\\ &y_5 = \varphi,\; y_6 = \varphi',\; y_7 = A_0,\; y_8 = A_r \end{aligned} \tag{59}$$

with the cosmological drift $\dot{\bar{\phi}}$ treated as an external constant on galactic timescales. The baryonic density profile $\rho(r)$ is prescribed (e.g. Plummer sphere for the proof-of-concept; exponential disk for SPARC comparison).

c. Boundary conditions. Regularity at $r = 0$ requires:

$$\Phi'(0) = \Psi'(0) = \varphi'(0) = A_r(0) = 0 \tag{60}$$

with $A_0(0)$ fixed by the constraint. Asymptotic flatness imposes:

$$\Phi(\infty) = \Psi(\infty) = \varphi(\infty) = A_r(\infty) = 0, \quad A_0(\infty)\to -1 \tag{61}$$

d. Continuation strategy. To ensure convergence of the nonlinear solver, we introduce a continuation parameter $\epsilon \in [0,1]$ multiplying the coupling, $\mathcal{F}\to\epsilon\mathcal{F}$, starting from the GR seed solution at $\epsilon = 0$ and increasing ϵ adiabatically to $\epsilon = 1$.

e. Outputs. From the converged solution we extract:

$$g_{\rm obs}(r) = \Phi'(r), \qquad g_{\rm bar}(r) = \frac{GM(r)}{r^2} \tag{62}$$

TABLE III. Status of the galactic field equations in the CRM vector sector. Four levels of epistemic confidence are distinguished. "Conditional" marks a result computed under an assumed density-dependent effective mass $m_{\rm eff}(\rho)$; that assumption is not supplied by the quadratic $f(R)$ sector, and its status is discussed in Sec. VI A.

Category	**Result**
Established: Existence	Coupled scalar-vector system admits static spherical solutions (Sec. VII)
Conditional: Scaling	Deep-MOND $\sqrt{g_{\rm bar}a_0}$ attractor via Chameleon feedback (Sec. VIII B)
Established: a_0	$a_0 = cH_0/(2\pi)$ as consistency relation (0.66σ from observed value)
Plausibilized: RAR shape	BVP extraction yields $\mu_{\rm CRM}(x) = [1-\exp(-x^\alpha)]^{-1}$, $\alpha = 0.310$
Plausibilized: SPARC	$\chi^2/{\rm dof} = 9.67$ using McGaugh proxy
Open: Exact $\mu(x)$	Analytical closed-form interpolation function from field equations
Open: Full SPARC	Reanalysis with CRM-native $\mu_{\rm CRM}$ instead of McGaugh proxy

and reconstruct the emergent interpolation function:

$$\mu\left(\frac{g_{\rm obs}}{a_0}\right) = \frac{g_{\rm bar}}{g_{\rm obs}} \tag{63}$$

as well as the RAR $g_{\rm obs}(g_{\rm bar})$. The deep-MOND attractor corresponds to $g_{\rm obs} \propto \sqrt{g_{\rm bar}\, a_0}$ in the low-acceleration regime.

B. Numerical BVP results: MOND attractor from operator feedback

We have implemented the nonlinear BVP described above using a Picard iteration scheme with a sparse tridiagonal solver and adiabatic λ-continuation. The key numerical finding is:

a. Operator feedback mechanism. The direct vector-induced acceleration Ξ computed from Eq. (48) using the linearized scalar gradient is numerically subdominant ($\Xi/a_0 \sim 10^{-47}$) due to the astronomical normalization $\rho_{\rm crit} \cdot r_s \cdot c^2 \sim 10^{36}$. The MOND enhancement arises instead through *operator-level feedback*: the effective Chameleon mass $m_{\rm eff}$ in the scalar equation depends on the local gravitational acceleration through the screening function:

$$m_{\rm eff}^2(r) = m_{\rm base}^2(r) \cdot f_{\rm screen}(g_{\rm obs}(r)) \tag{64}$$

where $f_{\rm screen}(g) = [1+(a_0/g)^\eta]^{-1}$ transitions from $f \to 1$ (screened, short Compton) at $g \gg a_0$ to $f \to (g/a_0)^\eta \ll 1$ (unscreened, long Compton) at $g \ll a_0$. This creates a closed feedback loop:

$$\varphi' \;\to\; \Xi \;\to\; g_{\rm obs} \;\to\; f_{\rm screen}(g_{\rm obs}) \;\to\; m_{\rm eff} \;\to\; \varphi' \tag{65}$$

b. MOND attractor at slope $= 0.5$. For a Plummer sphere ($M = 5\times10^{10}\,M_\odot$, $r_s = 3\,$kpc), the self-consistent Picard iteration converges to a RAR slope of 0.500 ± 0.001 across a *continuous* family of screening parameters $(\eta, n_{\rm grad})$:

- $(\eta, n_{\rm grad}) = (0.30, 0.28)$: slope $= 0.5003$, $V_{\rm flat} = 154\,$km/s, enhancement $2.9\times$.
- $(\eta, n_{\rm grad}) = (0.55, 0.30)$: slope $= 0.501$.
- $(\eta, n_{\rm grad}) = (0.80, 0.34)$: slope $= 0.500$.

The MOND attractor is *not fine-tuned*: it forms a one-dimensional curve in the two-dimensional $(\eta, n_{\rm grad})$ parameter space. All points on this curve produce the same deep-MOND phenomenology ($g_{\rm obs} \propto \sqrt{g_{\rm bar}\, a_0}$, flat rotation curves, Tully-Fisher relation).

c. Multi-galaxy BVP scan. To test whether the MOND attractor persists across a range of galaxy masses, we solve the BVP for six Plummer spheres spanning $M = 10^9$–$10^{12}\,M_\odot$:

- $M = 10^9\,M_\odot$: slope $= 0.575$, $V_{\rm flat} = 36.0\,$km/s, enhancement $4.9\times$.
- $M = 10^{9.5}\,M_\odot$: slope $= 0.552$, $V_{\rm flat} = 50.1\,$km/s, enhancement $3.8\times$.
- $M = 10^{10}\,M_\odot$: slope $= 0.556$, $V_{\rm flat} = 69.5\,$km/s, enhancement $3.0\times$.
- $M = 10^{10.5}\,M_\odot$: slope $= 0.585$, $V_{\rm flat} = 94.5\,$km/s, enhancement $2.3\times$.
- $M = 10^{11}\,M_\odot$: slope $= 0.643$, $V_{\rm flat} = 128.7\,$km/s, enhancement $1.7\times$.
- $M = 10^{12}\,M_\odot$: slope $= 0.719$, $V_{\rm flat} = 280.3\,$km/s, enhancement $1.2\times$.

The median RAR slope is 0.58. Low-mass galaxies ($M \lesssim 10^{10.5}\,M_\odot$) are deep in the MOND regime and cluster near slope ≈ 0.56, close to the ideal value of 0.5. More massive systems ($M \gtrsim 10^{11}\,M_\odot$) show steeper slopes (0.64–0.72) because a larger fraction of their rotation curve lies in the Newtonian regime ($g > a_0$), where the effective slope increases toward unity. This mass dependence is a *prediction* of the CRM: the deep-MOND attractor is cleanest for low-surface-brightness dwarfs and weakens progressively for massive spirals.

d. Physical interpretation. The self-regulation mechanism is: when $g_{\rm obs} < a_0$, the screening weakens ($f_{\rm screen} \to 0$), reducing $m_{\rm eff}$ and extending the scalar Compton wavelength. This allows φ' to persist at larger radii, increasing Ξ and thereby $g_{\rm obs}$. The system reaches equilibrium when $g_{\rm obs} \sim \sqrt{g_N \cdot a_0}$ – the MOND attractor. Conversely, when $g_{\rm obs} > a_0$, screening

strengthens, shortening the Compton wavelength and reducing Ξ, restoring Newtonian gravity. The attractor at slope = 0.5 is thus a structural fixed point of the Chameleon feedback, not an input assumption.

e. Extended scan and $\mu(x)$ extraction. A finer scan with 19 Plummer spheres spanning $M = 10^8$–$10^{12.5}\,M_\odot$ (in 0.25-dex steps) confirms the mass-dependent slopes. The emergent interpolation function $\mu_{\rm CRM}(x) = [1 - \exp(-x^\alpha)]^{-1}$ with $\alpha = 0.310 \pm 0.001$ (Eq. 58) provides an 11× better parametric fit to the binned BVP data than the McGaugh form (see Sec. VII H).

C. Zero-parameter SPARC test

We propose a definitive test using the SPARC database [16] of 175 galaxies with measured rotation curves and surface photometry.

Method:

1. Fix $a_0 = cH_0/(2\pi)$ using the Planck value $H_0 = 67.36 \pm 0.54\,\mathrm{km/s/Mpc}$. This is *not* a free parameter.
2. For each galaxy, the stellar mass-to-light ratio Υ_* is allowed to vary within astrophysical priors (± 0.2 dex).
3. Compute $g_{\rm CRM}(g_{\rm bar}, a_0)$ using the RAR formula (22).
4. Compare with observed $g_{\rm obs}$.

Likelihood:

$$\chi^2_{\rm SPARC} = \sum_{i=1}^{N_{\rm data}} \frac{(g_{{\rm obs},i} - g_{\rm CRM}(g_{{\rm bar},i},\, a_0))^2}{\sigma_i^2} \qquad (66)$$

MCMC configuration:

- Algorithm: `emcee` affine-invariant ensemble sampler [21]
- Walkers: 48
- Steps: 10,000 (after 1,000 burn-in)
- Convergence: autocorrelation time $\tau < 50$

Priors (from Paper III):

- $\alpha_{M,0} = 0.0011^{+0.0010}_{-0.0006}$
- $n = 0.55^{+0.58}_{-0.29}$
- $H_0 = 67.36 \pm 0.54\,\mathrm{km/s/Mpc}$
- Υ_*: free per galaxy (within ± 0.2 dex)

Smoking-gun test: We run two MCMC analyses:

- **Run A:** Standard MOND (a_0 free). Expected result: $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m/s^2}$.
- **Run B:** CRM-fixed ($a_0 = cH_0/(2\pi)$, fixed). If Run B achieves χ^2 values comparable to Run A (within, say, $\Delta\chi^2 \lesssim 2N_{\rm gal}$), this supports the consistency of the CRM-deduced value with galactic data—but does *not* constitute an independent prediction, since $a_0 = cH_0/(2\pi)$ is constructed to reproduce the observed a_0 (Sec. IV C). Significantly worse χ^2 would falsify this consistency relation.

Clarification and caveat on nomenclature: "Zero-parameter" refers to the absence of free dark matter halo parameters or MOND acceleration parameters. The per-galaxy Υ_* values (one per galaxy, 171 galaxies = 171 additional parameters) are astrophysical nuisance parameters present in *any* rotation curve analysis (including ΛCDM with NFW halos). However, readers should be aware that this label is potentially misleading: Run A has 171 + 1 free parameters (Υ_* per galaxy plus a_0), while Run B has 171 free parameters. The comparison tests only whether fixing a_0 to the CRM-predicted value degrades the fit, not whether the CRM is a genuinely parameter-free theory.

D. Full SPARC database test

We test the CRM prediction $a_0 = cH_0/(2\pi) = 1.042 \times 10^{-10}\,\mathrm{m/s^2}$ against the full SPARC database [16], using 171 galaxies with sufficient data quality (of 175 total). For each galaxy, the baryonic acceleration $g_{\rm bar}$ is computed from the stellar and gas mass profiles using the McGaugh interpolation (22). We perform three runs as defined in the test protocol (Sec. VIII C).

a. Results.

- **Run A** (free a_0 per galaxy): $\chi^2/\mathrm{dof} = 4.88$ ($N_{\rm dof} = 3033$), median $a_0 = 0.733 \times 10^{-10}\,\mathrm{m/s^2}$.
- **Run B** (CRM: $a_0 = cH_0/(2\pi)$ fixed): $\chi^2/\mathrm{dof} = 9.67$ ($N_{\rm dof} = 3204$).
- **Run C** (best global a_0, free): $\chi^2/\mathrm{dof} = 9.30$ ($N_{\rm dof} = 3204$), best-fit $a_0 = 0.880 \times 10^{-10}\,\mathrm{m/s^2}$.

The elevated χ^2/dof values reflect the fact that we use the empirical McGaugh interpolation (22) rather than the CRM-native interpolation function (which remains to be extracted from the nonlinear BVP). The McGaugh function is known to provide only an approximate description of the full SPARC scatter [31]. The $\Delta\chi^2$ between Run B (CRM-fixed) and Run C (best global) is 1183.

Critical assessment of fit quality: The factor-of-∼ 2 increase in χ^2/dof from Run A (4.88) to Run B (9.67) is statistically substantial and cannot be dismissed as a "modest penalty." Taken at face value, the data disfavor the CRM-fixed value $a_0 = cH_0/(2\pi) = 1.042 \times 10^{-10}\,\mathrm{m/s^2}$ over both the per-galaxy-free value (Run A median: 0.733) and the best global constant (Run C: 0.880). This is a genuine tension, not resolved by noting

that Runs B and C share the same $N_{\rm dof}$. The CRM-deduced a_0 lies $\approx 15\%$ above the best-fit global value; at SPARC's precision this is a statistically significant discrepancy. The full interpretation depends on three open issues: (i) the CRM-native interpolation function (unavailable until the BVP is solved analytically), (ii) the value of the saturation factor $\mathcal{B}_0$ (not yet constrained independently), and (iii) the possible systematic offset induced by using the McGaugh form for a model that predicts a different transition shape. The SPARC test as reported here is therefore an *exploratory consistency check*, not a confirmed validation.

The best-fit global $a_0 = 0.880\times10^{-10}\,\mathrm{m/s^2}$ lies *between* the CRM prediction (1.042) and the per-galaxy median (0.733), suggesting that the true transition scale may be shifted by the saturation factor $\mathcal{B}_0$ (see Assessment below).

b. Baryonic Tully-Fisher relation. The CRM prediction $V_{\rm flat}^4 = G\,M\,a_0^{\rm CRM}$ yields velocities that are 3.5% lower than the standard MOND prediction ($V_{\rm CRM}/V_{\rm MOND} = (a_0^{\rm CRM}/a_0^{\rm obs})^{1/4} = 0.965$), well within the observational scatter of the BTFR [6].

c. Assessment. The $\sim$ 13% discrepancy between $a_0^{\rm CRM} = cH_0/(2\pi)$ and the empirical value $a_0^{\rm obs} = 1.2\times10^{-10}\,\mathrm{m/s^2}$ is only 0.66σ given the total (statistical + systematic) uncertainty of $\sigma_{\rm tot} = 0.24\times10^{-10}\,\mathrm{m/s^2}$ [5]. The discrepancy admits several interpretations: (i) the saturation factor $\mathcal{B}_0 = \mathrm{sech}^2(\bar\phi/\phi_0)$ at the present epoch may deviate from unity, with $\mathcal{B}_0 \approx 0.87$ sufficient to close the gap; (ii) the emergent interpolation function from the nonlinear BVP (Sec. VIII B) may differ from the McGaugh form (22), shifting the effective transition scale; (iii) adopting the SH0ES value $H_0 = 73.04\,\mathrm{km/s/Mpc}$ [30] yields $a_0 = 1.13\times10^{-10}\,\mathrm{m/s^2}$, reducing the discrepancy to 6% (0.3σ). The a_0 prediction thus provides a novel connection to the Hubble tension.

Gupta and Samaras [17] provide a separate, model-dependent comparison of CCC, MOND and NFW fits using the 175-galaxy SPARC catalogue; their principal rotation-curve fits use 165 galaxies. In the 91-galaxy subset with resolved transition radii, Table 6 reports a median CCC transition acceleration $a_t = 2.1\times10^{-10}\,\mathrm{m/s^2}$ with 0.33 dex scatter, compared with a median fitted MOND scale $9.5\times10^{-11}\,\mathrm{m/s^2}$ and 0.34 dex scatter. These similar dispersions do not identify the two acceleration scales and do not derive a universal $a_0 = cH_0/(2\pi)$. The study is an empirical comparison under its sample-selection and fitting assumptions, not a validation of the CRM coupling function.

E. Predictions and falsifiability

Table IV lists testable predictions of the vector sector.

The model is **falsifiable**: if the SPARC MCMC shows that $a_0 = cH_0/(2\pi)$ produces a significantly worse fit than free a_0, the deductive connection between cosmological expansion and galactic dynamics is ruled out. Similarly, if pure dark-matter halos (without baryonic counterparts) are discovered, the nested hierarchy picture is falsified.

IX. DISCUSSION

A. Comparison with AeST

The Aether Scalar Tensor (AeST) theory of Skordis & Złośnik [7] is the closest existing framework to the CRM vector sector. Both theories employ a timelike unit vector field to generate MOND phenomenology within a relativistic framework. The key differences are:

1. **Origin:** AeST postulates the vector field; CRM derives it from thermodynamic hierarchy.
2. a_0**:** In AeST, a_0 is a free parameter; in CRM, $a_0 = cH_0/(2\pi)$ is deduced from the scalar-vector coupling.
3. **Dark energy:** AeST requires a cosmological constant; CRM replaces Λ with the Pöschl-Teller saturation of the scalaron.
4. **CMB compatibility:** AeST was demonstrated to fit the CMB power spectrum [7]; CRM has demonstrated CMB compatibility for the scalar sector (Paper III) but the vector contribution to perturbations remains to be computed.

A detailed comparison of the perturbation spectra (CMB C_ℓ and matter $P(k)$) between CRM and AeST would be highly informative but lies beyond the scope of this paper. Recent work has demonstrated that relativistic MOND theories with timelike vector fields – including AeST – can be reformulated as mimetic gravity theories, exposing deep conformal and disformal symmetries in the metric structure [35]. Whether the CRM vector sector admits a similar mimetic reformulation, potentially simplifying its stability analysis, is an open question.

B. Independent convergence: Imprint-Fading theory

Independently and by a different method, Kovacs [34] derives the same fundamental scale $a_0 = cH_0/(2\pi)$ from the equality of the Gibbons–Hawking temperature of de Sitter spacetime and the Unruh temperature at acceleration a_0. His "Imprint-Fading" (IF) theory introduces a scalar curvature-occupation field $s(\mathbf{x},t) \in (0,1)$ governed by a single dissipative evolution equation $\dot s = H_0[y - s^2/(1-s)]$, where $y \equiv g_{\rm N}/a_0$. In steady state, this yields the MOND simple interpolation function $\nu(y) = [1+\sqrt{1+4/y}]/2$ *exactly and algebraically* – a result that is not fitted but follows from the fixed-point condition. Global asymptotic stability is proved via a Lyapunov functional.

TABLE IV. Predictions and consistency checks of the CRM vector sector. Items marked † are *consistency relations* (the model is constructed to reproduce the observed value), not parameter-free predictions; see Sec. IV C, Epistemic caveat.

Item	Type	Test	Status
$a_0 = cH_0/(2\pi)$†	Consistency relation	SPARC MCMC	This paper
Flat rotation curves	Prediction	SPARC	This paper
Baryonic Tully-Fisher	Prediction	BTFR data	This paper
RAR universality	Prediction	SPARC	This paper
No DM halos w/o baryons	Prediction	LSST, SKA	Future
MOND weaker at high z	Optional prediction	JWST, ELT	Future
$\alpha_T = 0$ (preserved)	Consistency check	GW170817	Proven (Sec. VI D)
$\rho_A = 0$ on FLRW	Consistency check	Cosmological fits	Proven (Sec. VI E)
Parasitic screening	Consistency check	Solar System	Proven (Sec. VI B)

The convergence of two independent frameworks on the same a_0 relation – derived here from the saturation ODE attractor and Fourier frequency conversion (Sec. IV C), and by Kovacs from thermodynamic temperature equality – strengthens the case for its physical significance. Three further points of comparison deserve emphasis:

1. **Complementary MOND derivations:** The IF theory obtains MOND as an exact algebraic equilibrium of a one-field dissipative equation; the CRM obtains it as an emergent numerical attractor of the two-field (scalar + vector) system via BVP extraction. Both routes converge on MOND phenomenology but from distinct mathematical structures.

2. **Non-equilibrium predictions:** The IF theory provides a quantitative relaxation spectrum ($\tau = 0.09$–$204\,\mathrm{Gyr}$) and predicts hyperbolic halo decay after baryon stripping, with a sharp dynamical bifurcation at $s_0 = \ln 2$. These non-equilibrium predictions – testable with JWST, Gaia DR4, and WEAVE – have no analogue in the CRM or in any prior MOND formulation.

3. **Dark energy equation of state:** The IF framework predicts $w \geq -1$ as a consequence of the null energy condition applied to the Fisher information metric on $s \in (0, 1)$. This is directionally consistent with the DESI DR2 preference for dynamical dark energy in the $w_0 > -1$, $w_a < 0$ quadrant [29] and with the CRM Lagrangian: the $f(R) = R + 2\gamma R^2$ sector of Paper III generically yields $w \to -1$ from above, not from below. The phantom value $w = -1.355$ reported in Paper I is an artefact of the phenomenological tanh fit to Pantheon+ data with diagonal covariance and should not be regarded as a prediction of the Lagrangian theory. We therefore regard $w \geq -1$ as the natural CRM prediction, in agreement with the IF result.

The key difference in scope remains: the IF theory is currently developed at the non-relativistic galactic level (with a candidate covariant extension), while the CRM provides a complete cosmological framework validated against CMB, BAO, and supernova data. A synthesis of the IF thermodynamic structure with the CRM Lagrangian formulation may prove fruitful.

C. The coupling function: derivation vs. ansatz

We emphasize that the coupling function $\mathcal{F}$ in Eq. (12) is a *minimal-coupling ansatz* consistent with six constraints, not a unique deductive result. The three-factor structure (gatekeeper × saturation × projection) is physically motivated and constraint-minimal, but alternative forms satisfying the same constraints may exist.

The deductive chain of the CRM is:

- **Uniquely derived:** The need for a vector sector, its timelike character, the kinetic structure $F_{\mu\nu}F^{\mu\nu}$, the unit constraint.
- **Strongly constrained:** The coupling function $\mathcal{F}$ (6 conditions, specific functional form proposed but not unique).
- **From data:** K_B, γ, V_0, ϕ_0.

This is comparable to the status of the Starobinsky model: the R^2 term is the simplest ghost-free extension of GR, but $R^2 + \epsilon R^3$ cannot be excluded on purely theoretical grounds. The selection is made by Occam's razor and empirical fit.

Non-uniqueness of the vector sector. We emphasize that the vector sector presented here is a *minimal consistent realization* of the thermodynamic requirement, not a unique result. The thermodynamic argument (Sec. II) establishes that an additional degree of freedom beyond the scalaron is *necessary* and that it must be a timelike vector field (Sec. II B). However, the specific coupling $\mathcal{F}$ (Eq. 12) is the simplest ansatz satisfying six constraints – alternative forms with the same constraint compliance may exist. The non-uniqueness is threefold: (i) the coupling function is constrained but not derived uniquely; (ii) the sech^2 modulation is motivated by Pöschl-Teller consistency but could be replaced by other smooth decoupling functions; (iii) the kinetic structure $F_{\mu\nu}F^{\mu\nu}$ is minimal but not the only ghost-free option (cf. generalized Proca theories). What *is* unique

is the existence and scaling behavior: a vector degree of freedom coupling to the scalaron gradient, suppressed in the Solar System, and producing $a_0 \sim cH_0$. The precise functional form awaits further theoretical development.

Remark 1 (Status of galactic field equations) *The CRM field equations at the galactic scale have the following status:*

- ***Established:*** *Existence of the MOND-scale attractor and $\sqrt{g_N}$ scaling from the saturation ODE.*
- ***Plausible:*** *The specific functional form of the interpolation function $\mu(x)$ from BVP extraction.*
- ***Open:*** *Exact derivation of $\mu(x)$ from first principles without numerical fitting.*

D. Open theoretical questions

1. **RAR interpolation function from field equations:** The BVP extraction across 19 galaxies (10^8–$10^{12.5}\,M_\odot$) yields a CRM-native interpolation function $\mu_{\rm CRM}(x) = [1 - \exp(-x^\alpha)]^{-1}$ with $\alpha = 0.310 \pm 0.001$ (Eq. 58). This differs from the McGaugh form ($\alpha = 0.5$) and provides an $11\times$ better fit to the BVP data ($\chi^2_{\rm red} = 1{,}467$ vs. $16{,}519$). The remaining task is a full SPARC reanalysis using $\mu_{\rm CRM}$ instead of the McGaugh proxy, which may improve the χ^2/dof of Run B.

2. **Rigorous 2π derivation:** The factor 2π in $a_0 = cH_0/(2\pi)$ has been confirmed by three related perspectives (Sec. IV C): the saturation ODE attractor at $\mathcal{B}_0 = 1$, the Fourier angular-to-linear frequency conversion ($f_H = H_0/(2\pi)$), and the uniqueness of the dimensional fixed point $a_0/(cH_0) = 1/(2\pi)$. *Note: these perspectives share common premises (the CRM saturation framework) and should not be counted as fully independent derivations of a_0.* The 15.2% discrepancy with $a_0^{\rm obs}$ is 0.66σ (Planck H_0) or 0.29σ (SH0ES H_0). The 2π factor is thus the standard angular-to-linear frequency conversion, not a free parameter or heuristic.

3. **Uniqueness of $\mathcal{B}(\phi)$:** The sech^2 form is motivated by Pöschl-Teller consistency but not proven unique. An exclusion argument (analogous to the $R + \gamma R^2$ uniqueness in Paper III) is desirable.

4. **Vector perturbation theory:** A semi-analytical EFT estimate of the vector sector's contribution to the CMB power spectrum yields $\Delta C_\ell/C_\ell < 10^{-11}\%$, well below the percent-level threshold for Planck consistency [27]. The estimate uses the EFT parametrization $\alpha_{M,0} = 0.0011$, $\alpha_T = 0$, $\alpha_B = -\alpha_M$ with the parasitic screening function from Sec. VI B. Combined with $\rho_A = 0$ on FLRW (Sec. VI E), this confirms that the vector sector is invisible to current CMB observations. A full numerical computation with `hi_class` is therefore not required for Planck-level consistency but remains desirable for future precision tests.

5. **Perturbation stability and well-posedness:** Ghost freedom (Constraint 5) is necessary but not sufficient for a well-posed theory. A complete stability analysis requires: (i) the quadratic action for scalar and vector perturbations around FLRW, (ii) the dispersion relations for all propagating modes, (iii) positivity of c_s^2 (no gradient instabilities), (iv) subluminality or controlled characteristics (a known subtlety in Einstein-Aether theories [25]), and (v) absence of strong coupling in the deep-MOND regime. We have established $\alpha_T = 0$ for tensor modes (Sec. VI D), but the scalar and vector perturbation sectors remain to be analyzed for the specific coupling (12).

6. **Quantitative Baryogenesis:** Paper I argues qualitatively that the baryon asymmetry $\eta \sim 6 \times 10^{-10}$ follows from MEPP-optimal annihilation. A quantitative derivation showing that MEPP selects this specific value would elevate the argument from "consistent with" to "predicted by."

X. CONCLUSION

We have presented the complete Curvature Relaxation Model action $S_{\rm CRM} = S_{\rm T1} + S_{\rm T2}$, incorporating both the scalar sector (dark energy, cosmological dark matter effects) and the vector sector (galactic MOND dynamics). The key results are:

1. The vector sector is *derived* from thermodynamic principles: MEPP requires stable dissipative structures (galaxies), which the scalar sector alone cannot provide. Its field equations and energy-momentum tensor are explicitly constructed (Sec. III D).

2. The coupling function $\mathcal{F} = |T|/\rho_{\rm crit} \cdot \mathrm{sech}^2(\phi/\phi_0) \cdot A_\mu \partial^\mu \phi$ satisfies all six constraints (BBN, MOND limit, screening, $a_0 \sim cH_0$, ghost freedom, $\alpha_T = 0$).

3. The MOND scale $a_0 = cH_0/(2\pi)$ emerges as an order-of-magnitude prediction (within $\sim 13\%$, or 0.66σ, of the observed value), not a fitted parameter. Galaxies "know" the expansion rate because the vector field couples to the scalar field that drives the expansion. Three related perspectives (sharing common CRM premises) support the 2π factor: the saturation ODE attractor, the Fourier angular-to-linear frequency conversion, and the dimensional fixed point uniqueness (Sec. IV C); see note in that section regarding their shared foundations.

4. Three rigorous consistency proofs are established: (i) $\alpha_T = 0$ exactly, preserving $c_T = c$ (Sec. VI D); (ii) parasitic screening suppresses the vector source in the Solar System by a factor $\sim 10^{-3\times10^9}$ (Sec. VI B); (iii) $\rho_A = 0$ on FLRW, ensuring the vector sector does not affect cosmological background fits (Sec. VI E).

5. The static spherical limit is derived explicitly (Sec. VII): the quasi-static decomposition $\phi = \bar{\phi}(t) + \varphi(r)$ reveals that the cosmological drift $\dot{\bar{\phi}} \sim H_0\phi_0$ is the physical bridge connecting expansion to galactic dynamics. The linearized reduction yields $g_A \propto g_N$ (constant rescaling); the numerical BVP (Sec. VIII B) demonstrates that the full nonlinear Chameleon feedback produces a deep-MOND attractor with RAR slopes ≈ 0.55–0.58 for low-mass galaxies ($M \lesssim 10^{10.5}\,M_\odot$), steepening to ≈ 0.72 for massive systems ($M \sim 10^{12}\,M_\odot$). The mass-dependent slope is a prediction: dwarf galaxies reside deeper in the MOND regime and approach the ideal slope of 0.5 more closely. The attractor is realized through the acceleration-dependent effective mass (operator feedback) and is not fine-tuned but forms a continuous family in parameter space. These are results *about the computation*: the BVP was solved with a prescribed density- and acceleration-dependent effective mass. That such a mass follows from the CRM Lagrangian is precisely what is no longer claimed (Sec. VI A), so the attractor demonstrates that the assumed feedback structure suffices, not that the model supplies it.

6. A full test against 171 SPARC galaxies (Sec. VIII D) shows that $a_0 = cH_0/(2\pi)$ produces a fit with $\chi^2/\mathrm{dof} = 9.67$ (CRM-fixed) vs. 4.88 (free a_0 per galaxy). The elevated χ^2 values reflect the use of the empirical McGaugh interpolation rather than the CRM-native form. The $\sim 13\%$ offset from the observed $a_0 = 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ is only 0.66σ and may be attributable to the saturation factor $\mathcal{B}_0 \approx 0.87$ or to the Hubble tension (SH0ES H_0 reduces the discrepancy to 0.3σ). The baryonic Tully-Fisher relation is reproduced to 3.5% accuracy.

7. The 4/3 convergence factor from two independent derivations (perturbation theory and phase-space geometry) provides structural evidence for the nested hierarchy.

The ontological picture is: two axioms (null-space fluctuation + thermodynamic optimization) → two degrees of freedom (scalar + vector) → three phenomena (dark energy, cosmological dark matter effects, MOND-compatible galactic dynamics) → one termination condition (standard physics below galactic scales). No Λ, no dark matter particles – only geometry. The termination condition, however, is now the weakest link rather than a settled corner: it rests on a screening mechanism that the quadratic $f(R)$ sector does not deliver (Sec. VI A), so the return to standard physics at small scales is assumed rather than derived. The numerical BVP confirms that Chameleon feedback can produce a MOND-like attractor (median slope ≈ 0.58 across six decades of galaxy mass), while the exploratory 171-galaxy SPARC check also exposes a real tension: the CRM-fixed value $a_0 = cH_0/(2\pi)$ remains within the broad empirical a_0 uncertainty but fits substantially worse than free or best-global a_0 under the McGaugh proxy. The extraction of the emergent interpolation function $\mu_{\mathrm{CRM}}(x) = [1 - \exp(-x^\alpha)]^{-1}$ with $\alpha = 0.310 \pm 0.001$ from the BVP (Eq. 58) reveals that the CRM-native transition is shallower than the McGaugh form ($\alpha = 0.5$) and provides an $11\times$ better fit to the numerical data. A full SPARC reanalysis using μ_{CRM} is therefore required before the galactic sector can be regarded as validated. Semi-analytical EFT estimates confirm that vector perturbations contribute $\Delta C_\ell/C_\ell < 10^{-11}\%$ to the CMB, establishing Planck-level safety of the vector sector.

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